Integrated Study of 7-Hydroxy-4-methylcoumarin

From Pechmann synthesis to fluorescence properties.



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I - From design to synthesis

1) Background and challenges in coumarin synthesis [M. Franco-Pérez]

Medicinal applications:

- Anticoagulant properties
- Antifungal activity
- Antibacterial agents

Optical properties:

- Fluorescence under UV light
- Applications in materials science
- Medical imaging and diagnostics

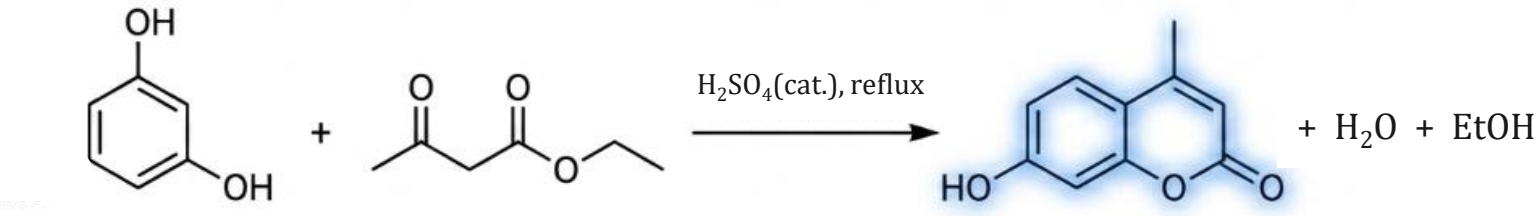
Project objectives:

- Develop an optimized synthesis suitable for teaching
- Short reaction time and readily available reagents
- Reaction monitoring by fluorescence

I - From design to synthesis

2) Organic synthesis strategy [M. Franco-Pérez]

Pechmann reaction (1883):



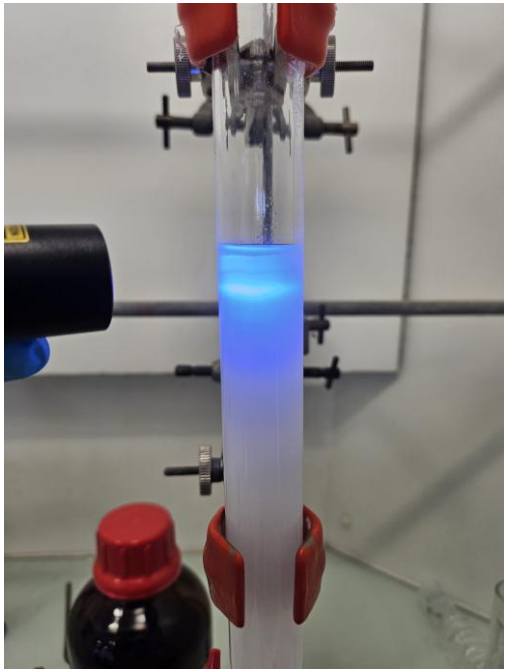
Resorcinol Ethyl acetoacetate 7-hydroxy-4-methylcoumarin

Experimental conditions:

Reactants: resorcinol and ethyl acetoacetate
Solvent: acetonitrile
Catalyst: H_2SO_4
Temperature: reflux at 80 °C

II - Experimental development and optimization

1) Chromatographic methods for reaction monitoring and purification



Column chromatography (~25 g silica)

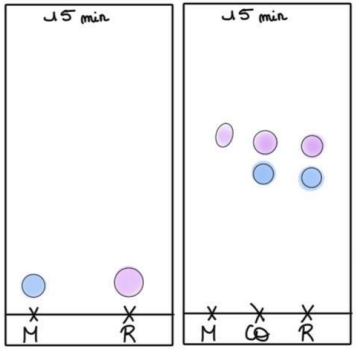
Stationary phase: silica
Mobile phase: cyclohexane / ethyl acetate
Detection: UV light

Reaction mixture:

- $Rf_{reaction\ 1}$ = 0.10
- $Rf_{reaction\ 2}$ = 0.46

Reference:

- $Rf_{reaction\ 1}$ = 0.13
- $Rf_{reaction\ 2}$ = 0.59



Legend :

M: reaction mixture

R: resorcinol

Co: Co-spot

Purple : Fluorescence quenching of the TLC plate

Blue : Fluorescence

TLC, reaction 1 left (4:1), reaction 2 right (1:1)

II - Experimental development and optimization

2) Optimization: synthesis 1 vs. synthesis 2

Parameter	Synthesis 1 (Initial)	Synthesis 2 (Optimized)
TLC eluent	Cyclohexane/EtOAc 80:20 → poor migration	Cyclohexane/EtOAc 50:50 → good separation
Reaction time	25 min → unnecessary	15 min → near-complete reaction
Column eluent	EtOAc → efficient elution	Cyclohexane/EtOAc 50:50 → band broadening
Column loading	Direct loading of crude oil → product loss	Pre-adsorption into silica → quantitative recovery
Yield	~ 72%	~84%

III – Structural validation of the coumarin derivative

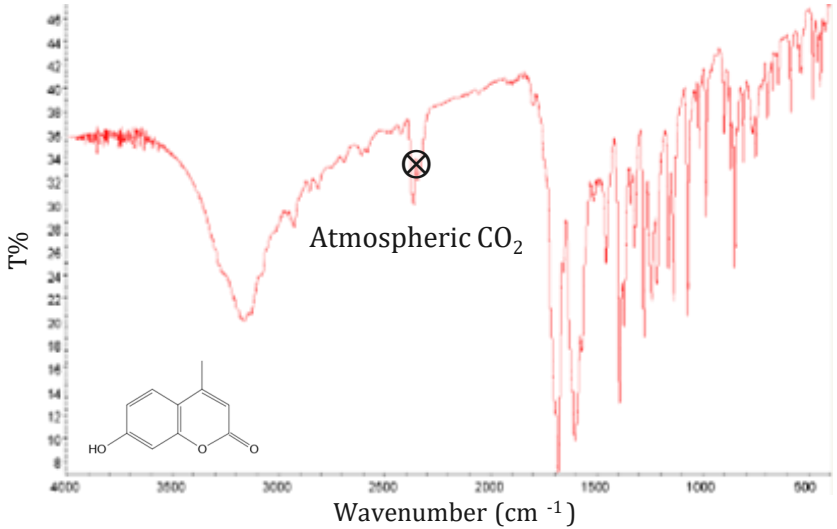
1) Characterization

Melting point (Kofler bench):

- $Mp_{experimental}$ ~185°C
- $Mp_{literature}$ = 185 -188°C

Functional group identification (FTIR):

- O–H stretching: broad band (~3200 cm^{-1})
- C=O stretching: strong, sharp band (~1700 cm^{-1}) conjugated ester
- C=C stretching: strong, sharp band (~1600 cm^{-1})

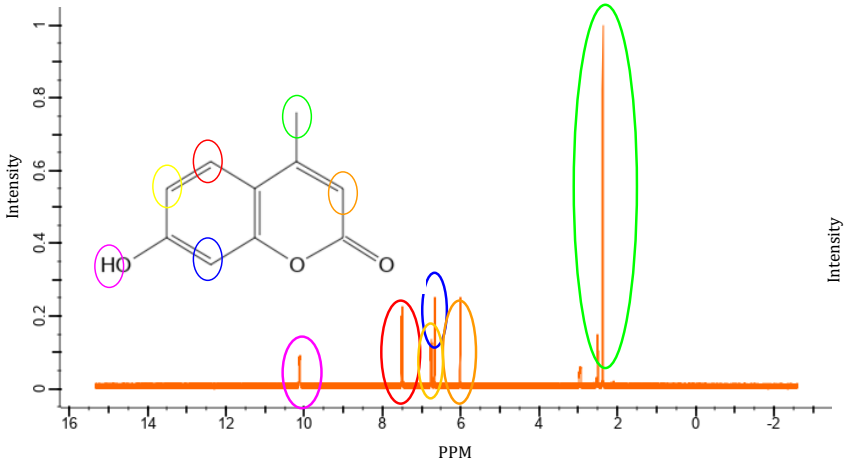


FTIR spectrum

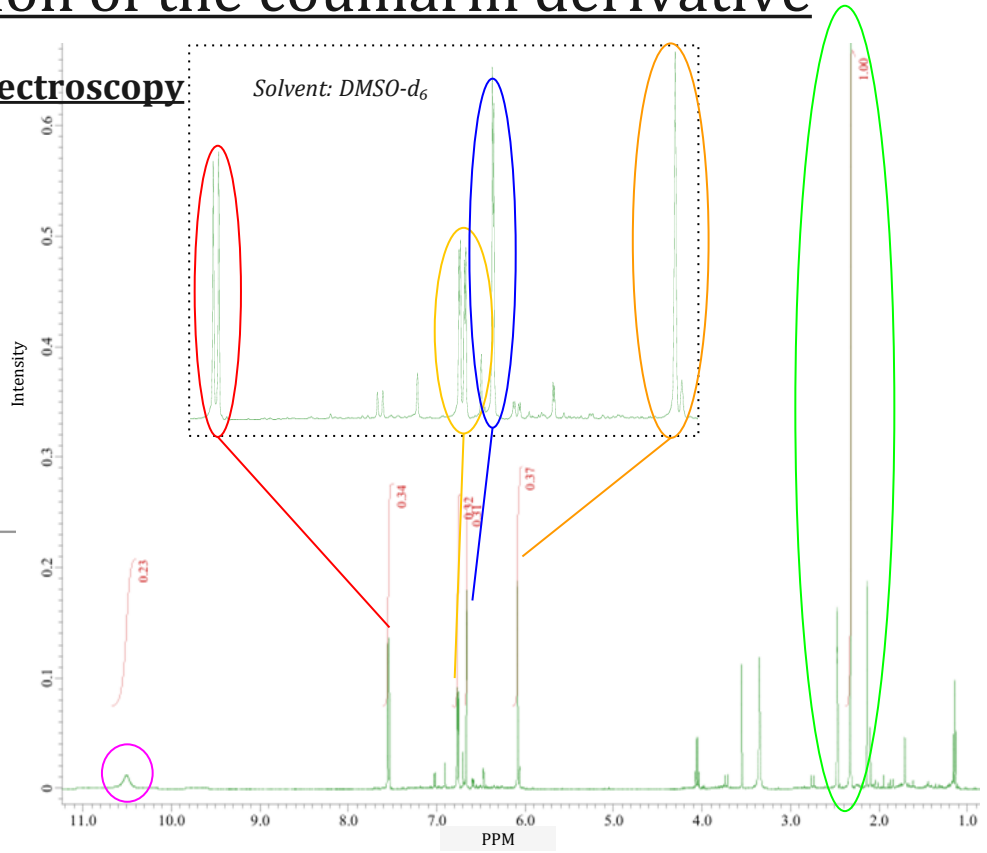
➔ Results consistent with the expected characteristics of 7-hydroxy-4-methylcoumarin

III – Structural validation of the coumarin derivative

2) Structure determination by ^{1}H NMR spectroscopy

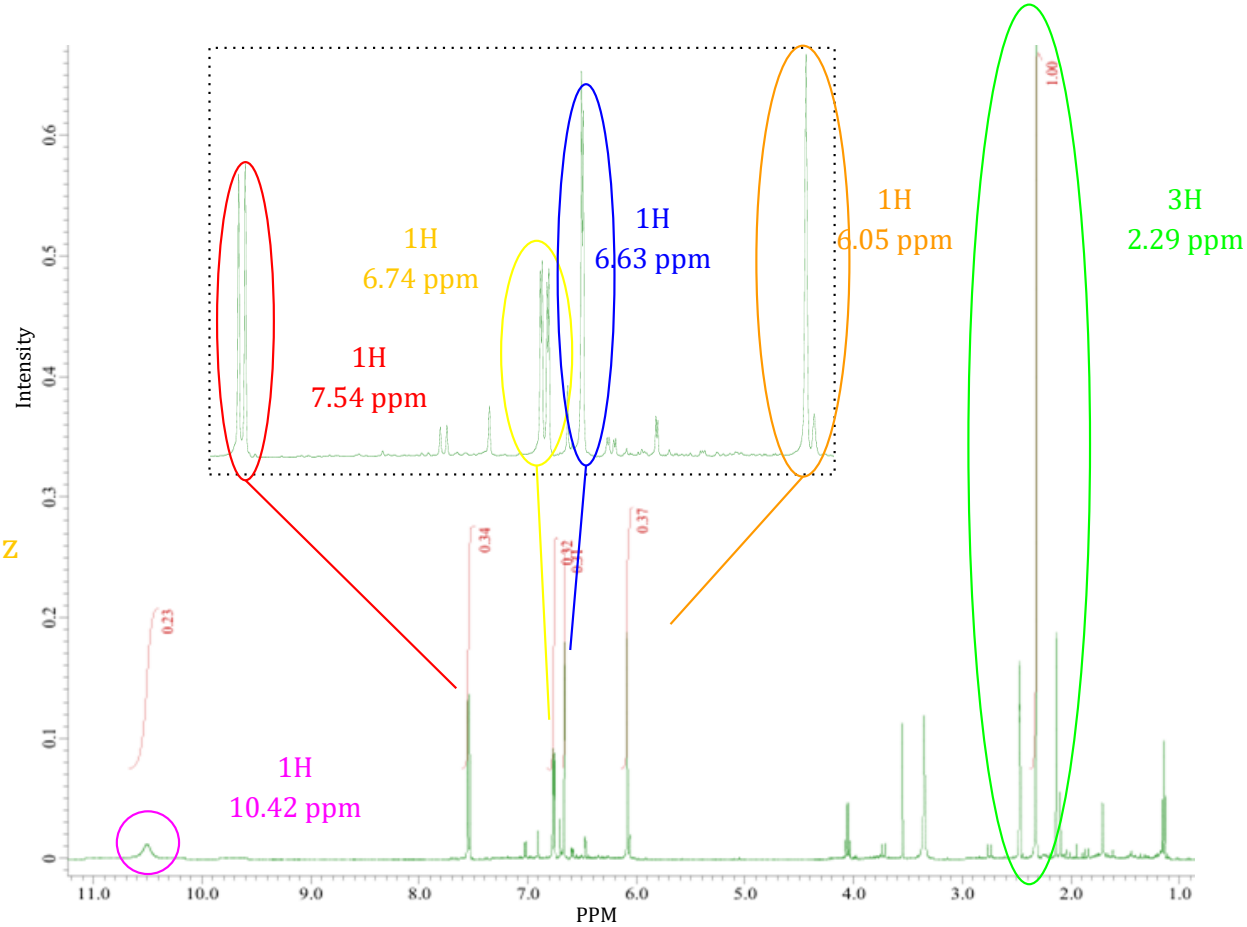


Reference ^{1}H NMR spectrum of 7-hydroxy-4-methylcoumarin [Sigma-Aldrich]



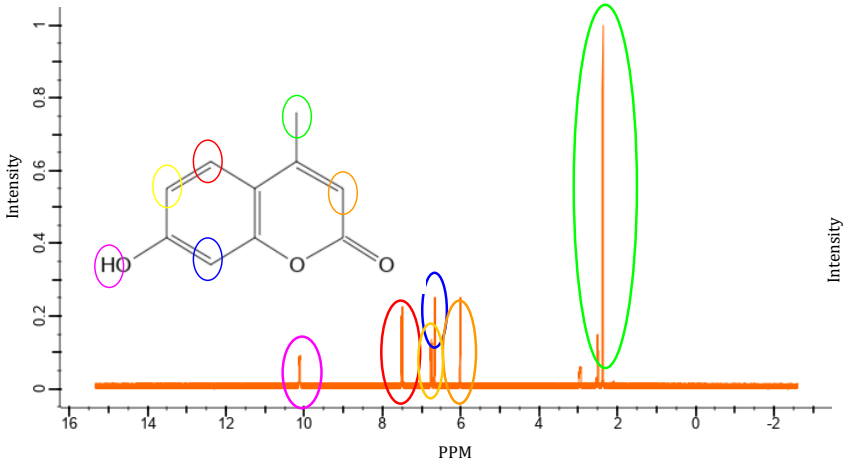
^{1}H NMR spectrum of 7-hydroxy-4-methylcoumarin

- Singlet
- Singlet
- Doublet : $J_{méta}$= 2.5 Hz
- Doublet of doublets :
- $J_{méta}$≃2.25 Hz, J_{ortho}≃ 8.75 Hz
- Doublet :
- J_{ortho}= 8.5 Hz
- Broad Singlet



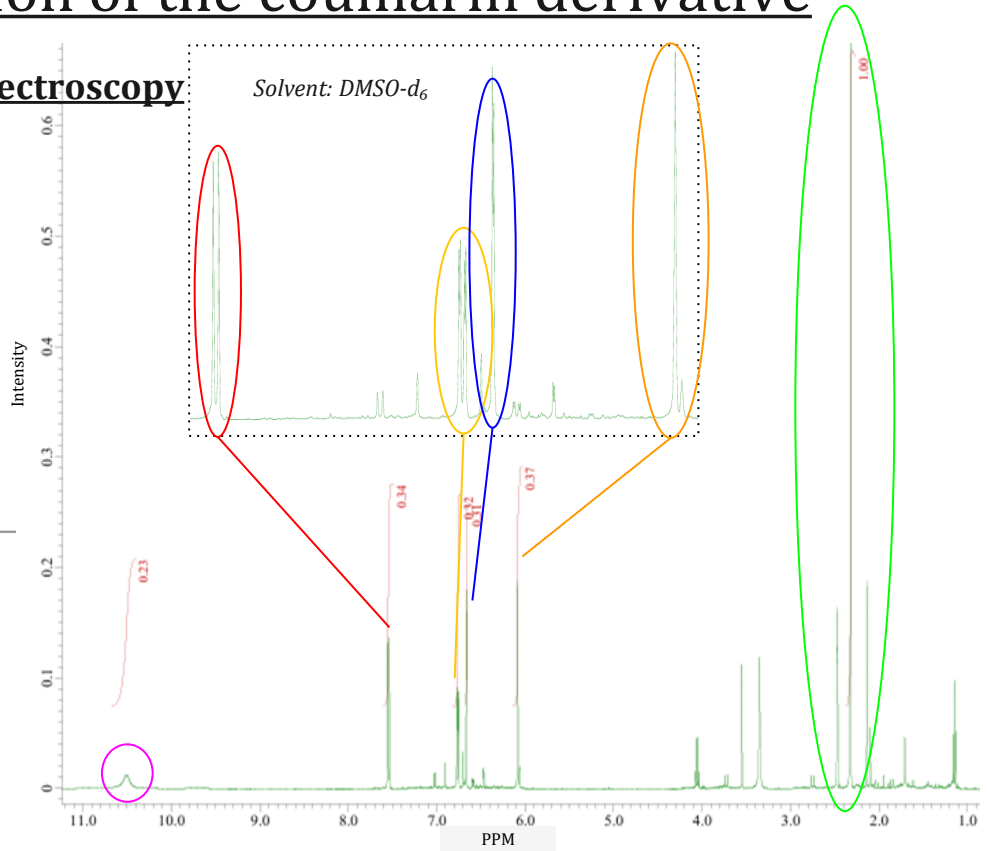
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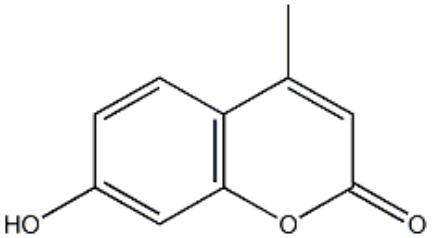
➔ Confirmation of the structure of 7-hydroxy-4-methylcoumarin



^{1}H NMR spectrum of 7-hydroxy-4-methylcoumarin

IV - Influence of pH on fluorescence

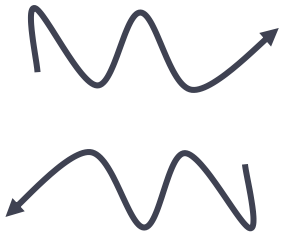
1) Fluorescence properties of the coumarin derivative [R.Koehler]



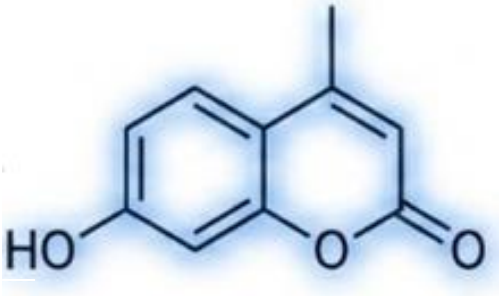
Ground state (S0):

- $\lambda_{ex} \simeq$ 330-370 nm
- Theoretical pKa $\simeq$ 7.3-7.8
- Experimental pKa $\simeq$ 7.67

+h**ν** (UV absorption)



relaxation → fluorescence emission



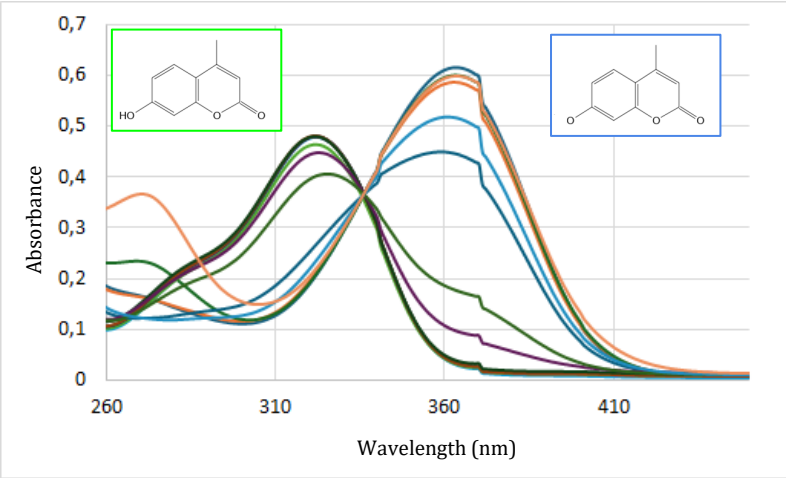
Excited state (S1) :

- λ_{em} = 450 nm (intense blue emission)
- pKa* ~ 1-2

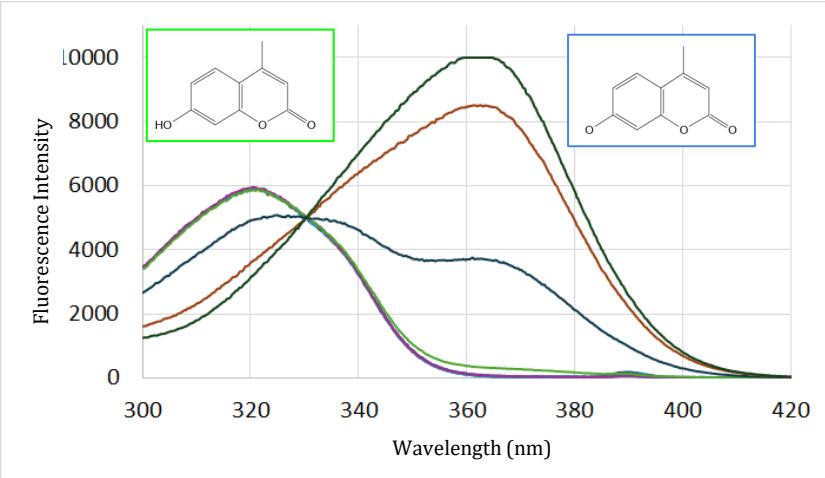
Stokes shift :
$\lambda_{em} > \lambda_{ex}$

IV - Influence of pH on fluorescence

2) Spectroscopic monitoring of the acid–base equilibrium



Experimental UV–Vis spectra of 7-hydroxy-4-methylcoumarin (C = 5.7 × 10^{-5} M), with pH variation

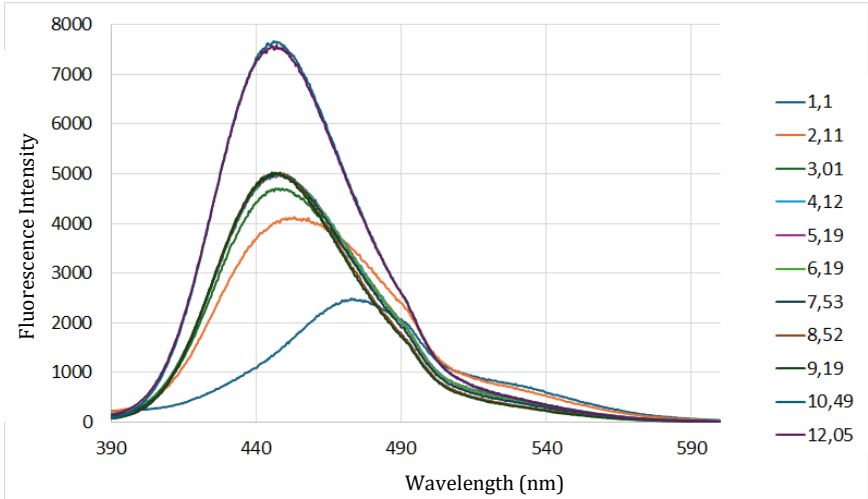


Fluorescence excitation spectra of 7-hydroxy-4-methylcoumarin (C = 5.7 × 10^{-5} M), emission wavelength fixed at 450 nm, with pH variation

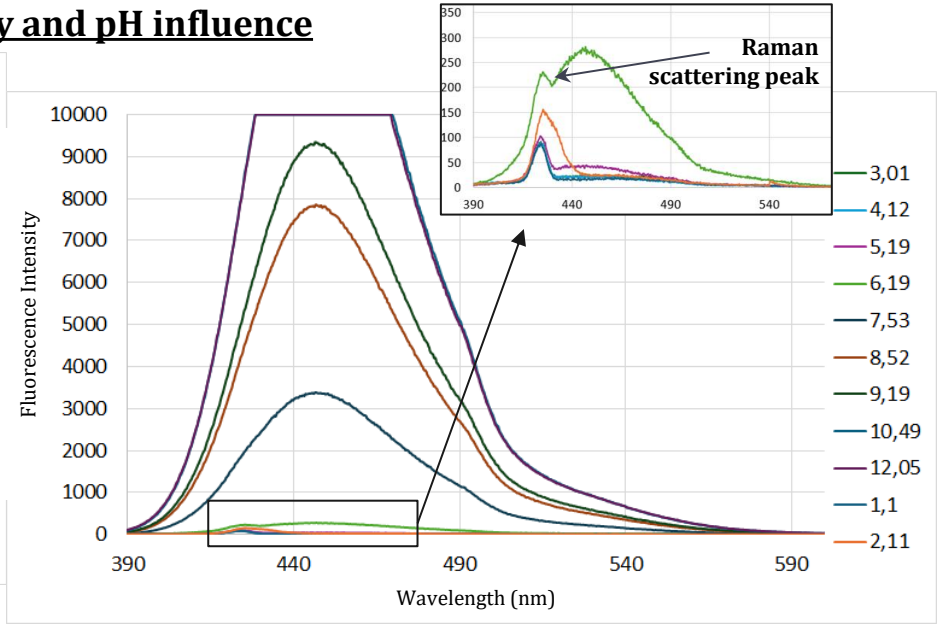
- **Acidic form absorption maximum:** 322 nm in UV–Vis, 321 nm in fluorescence excitation
- **Basic form absorption maximum**: 364 nm in UV–Vis, 361 nm in fluorescence excitation
- **Isosbestic/isoexcitation point**: 335 nm in UV–Vis, 330 nm in fluorescence excitation

IV - Influence of pH on fluorescence

3) Fluorescence emission spectrofluorimetry and pH influence



Fluorescence emission spectra of 7-hydroxy-4-methylcoumarin (C =5.7.10^{-7} M), excitation wavelength fixed at 330 nm, with pH variation



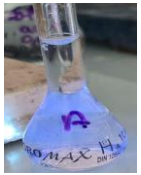
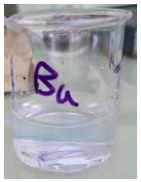
Fluorescence emission spectra of 7-hydroxy-4-methylcoumarin (C =5.7.10^{-7} M), excitation wavelength fixed at 370nm, with pH variation

- **Emission maximum** constant at ∼ **450 nm**
- Fluorescence intensity : $\Phi_{basic\ form} > \phi_{acidic\ form,}$ with $\Delta_{If}(370nm) > \Delta_{If}(330nm)$
- **Predominant emission from the anionic form**

IV - Influence of pH on fluorescence

4) Demonstration by Dykstra of the protective role of buffers

HCl addition → effect on coumarin fluorescence in different medium
Each sample contains 0,10 mL of the coumarin derivative (C_0=5.7.10^{-4} M)

HCl (0.1M)	A: Unbuffered (0.90 mL distilled water)	Bc: Carbonate buffer (0.90 mL, 0.05M, pH=9)	Bp: Phosphate buffer (0.90mL, 0.05M, pH=7.5)	Ba: Acetate buffer (0.90mL, 0.05M, pH=4.5)
No HCl added				
Small quantity of HCl added				
Large quantity of HCl added				

V – Conclusion and experimental recommendations

Optimal synthesis conditions : 15 minute reaction time, TLC monitoring using Cyclohexane/EtOAc (1:1), pre-adsorption into silica and AcOEt as eluent for column chromatography

Suggested improvement: Adjust the pH after the reaction to facilitate product characterization (reaction medium is acidic)

Demonstration of fluorescence properties : Use buffer solutions to easily demonstrate the fluorescence properties of the coumarin derivative and the buffering capacity during laboratory sessions

Perspectives :

- Incorporate spectrometric analysis into the experiment to provide an interdisciplinary approach → highlighting the connection between organic chemistry, solution chemistry, and spectroscopy
- Assess product purity using liquid chromatography coupled with UV–Visible detection

VI - Bibliography

[M. Franco-Pérez] : A Novel and Simple Experimental Procedure for Synthesizing, Purifying, and Identifying Coumarin in Undergraduate Heterocyclic Chemistry Laboratories. Mónica I. García-Aranda, Marco Franco-Pérez, and Angel Zamudio-Medina. Journal of Chemical Education 2023 100 (10), 4001-4006.

[J. Lalande et A. Parenty] : Synthèse d'une coumarine, l'actualité chimique, 2003.

[R.Koehler] : Fink, D. W., & Koehler, W. R. (1970). pH Effects on Fluorescence of Umbelliferone. Analytical Chemistry, 42(9), 990–993.

[Dykstra] : Dykstra, M., Chung, Y., & Muyskens, M. (2024). Easy, Bright, Fluorescence Demonstration of Buffer Action. Journal of Chemical Education, 101(2), 701–705.

[Sigma-Aldrich] : www.sigmaaldrich.com, Fiche de données de sécurité et spectres de référence RMN pour la 7-Hydroxy-4-methylcoumarine.

[Kürti - Czakó] : Kürti, L., Czakó, B. Strategic Applications of Named Reactions in Organic Synthesis. Elsevier, San Diego, 2005.

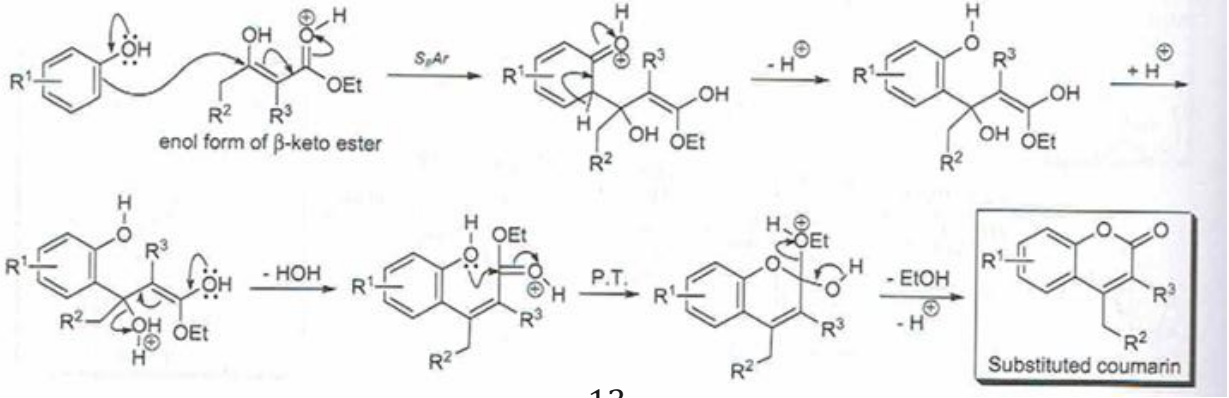
VII - Annex 1 : Reaction mechanism [Kürti - Czakó]

von Pechmann reaction:

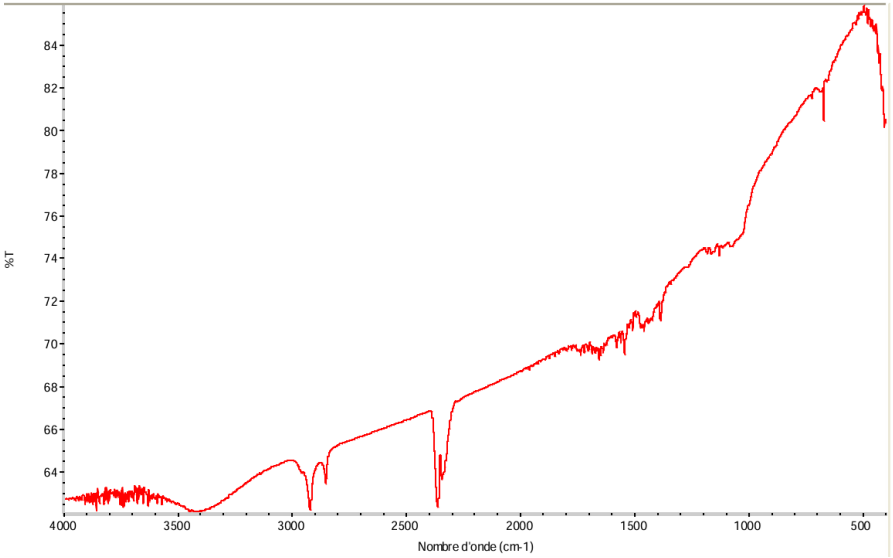


R^1 = H, OH, *O*-alkyl, NH_2, NHR, NR_2; R^2 = H, alkyl, aryl; R^3 = H, alkyl, aryl, Cl; protic acid: H_2SO_4, HCl, H_3PO_4; Lewis acid: $POCl_3$, $ZnCl_2$, $AlCl_3$, $FeCl_3$, $InCl_3$, $Yb(OTf)_3$, $SnCl_2$, $TiCl_4$, $SiCl_4$, PPA

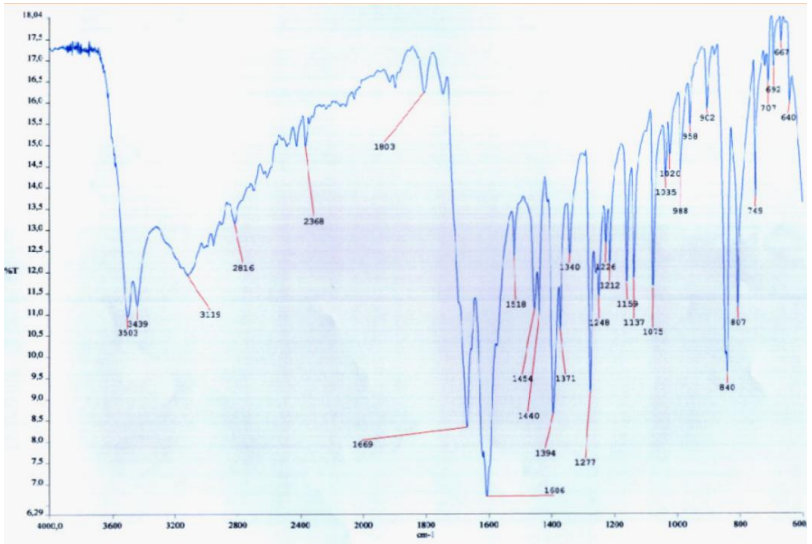
Mechanism: [28,29]



VII - Annex 2 : FTIR spectrum



Experimental FTIR spectrum of KBr



IR spectrum of the 7-hydroxy-4-méthylcoumarine [J. Lalande et A. Parenty]